

KELT-23Ab

R-2334

Benchmark run · workflow W8-benchmark-deep · record deep-bench_KELT-23A_250907 · created 2026-08-18T23:35:12+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460925.666 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.047
Transit depth (Rp/Rs)^2	2.7 +/- 1.54 [%]
Orbital Inclination (inc)	84.81 +/- 2.57
Airmass coefficient 1 (a1)	0.047 +/- 0.032
Airmass coefficient 2 (a2)	1.72 +/- 0.45
Scatter in the residuals of the lightcurve fit is	61.33 %
Best Comparison Star	#3 - [295, 262]
Optimal Aperture	2.64
Optimal Annulus	3.5
Transit Duration (day)	0.069 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.164 $\pm$ 0.047 vs 0.132 (deviation 0.25 σ)
Duration fit vs published	0.069 d vs 0.0954 d (deviation 0.37 σ)
Mid-time O-C	-1.7 min
Depth SNR	1.3
Residual scatter	61.33 %

Light curve

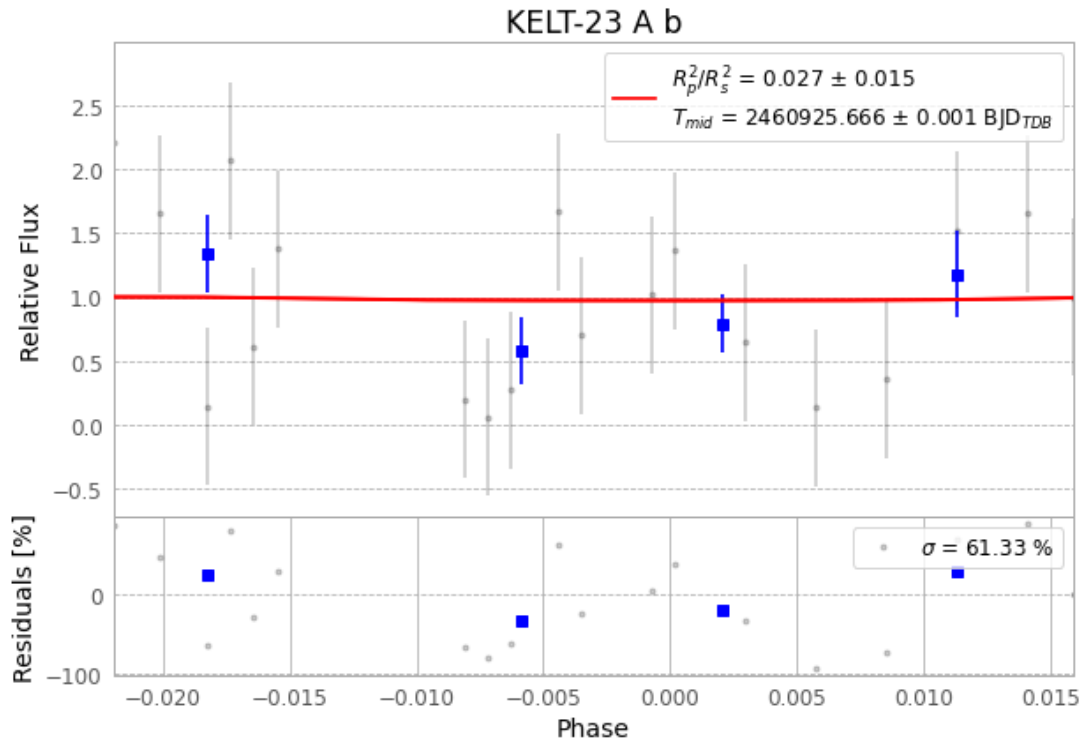


Figure 1 — FinalLightCurve_KELT-23 A b_2025-09-06.png (SHA-256 ec58da3928bf2d9e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [524, 279], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3bffa1bea0b80b03...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

42 FITS frames (28 MB), KELT-23A250907024815.FITS → KELT-23A250907045115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KELT-23A-250907.log (0edc61d43aec...), FinalLightCurve_KELT-23 A b_2025-09-06.png (ec58da3928bf...), FinalLightCurve_KELT-23 A b_2025-09-06.csv (d9065dde0dcf...)

Compute resources

Wall time 154.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 23:35Z.